STUDENT PERFORMANCE PREDICTION SYSTEM

Machine Learning Internship Project Report

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Project type	Beginner-level Machine Learning project
Technology stack	Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Streamlit
Dataset	UCI Student Performance — Mathematics dataset
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Submitted as an internship project deliverable. This report describes the implemented workflow and the evaluation results recorded during development.

Abstract

The Student Performance Prediction System applies supervised machine learning regression to estimate a student's final Mathematics grade (G3) from demographic, family, school, social and study-related attributes. The project includes data inspection, exploratory data analysis, preprocessing, train/test splitting, model training, evaluation and a Streamlit interface. Decision Tree and Random Forest regressors were compared. A separate early-prediction Random Forest model was trained without the period grades G1 and G2 and is used by the interactive application.

1. Problem Statement and Objectives

Student performance is influenced by multiple academic, personal and contextual factors. The objective of this project is to explore student data and build a regression-based system that estimates final grade G3. The prediction is intended as an analytical estimate, not as a definitive judgement of a student's ability.

- Inspect and understand the student dataset.
- Clean and analyze the data using descriptive statistics and visualizations.
- Prepare categorical and numerical features for machine learning.
- Train and compare regression models using a held-out test set.
- Evaluate models using MAE, RMSE and R^2.
- Build an interactive Streamlit application for early grade estimation.

2. Dataset and Data Understanding

The main analysis uses student-mat.csv from the UCI Student Performance dataset. The recorded dataset contains 395 rows and 33 columns. The target variable is G3, the final Mathematics grade, and the initial feature set contains 32 predictors. The project folder also contains student-por.csv, the Portuguese-language dataset, but the reported model results in this report refer to the Mathematics dataset.

During the recorded inspection, no missing values and no duplicate rows were found. The dataset includes numerical and categorical attributes. Categorical variables were identified separately from numerical variables for preprocessing.

3. Exploratory Data Analysis

Exploratory analysis included descriptive statistics, a histogram of final grades and a correlation heatmap. The recorded correlations showed strong positive relationships between G3 and the period grades G1 (approximately 0.80) and G2 (approximately 0.90). The correlation between past failures and G3 was approximately -0.36. Correlation indicates association and does not establish causation.

The following chart files were generated in the project’s visualizations folder: grade_distribution.png, correlation_heatmap.png, model_comparison.png and feature_importance.png. They should be retained alongside this report in the final repository.

4. Data Preprocessing and Feature Engineering

The target G3 was separated from the predictors. Categorical features were transformed using OneHotEncoder(handle_unknown='ignore'), while numerical features were passed through unchanged. These transformations were combined using a Scikit-learn ColumnTransformer and placed inside model Pipelines so that preprocessing is applied consistently during training and prediction.

For the full model, the dataset was split into training and testing subsets using an 80:20 ratio and random_state=42. This produced 316 training records and 79 test records. The early-prediction model excludes G1 and G2 from its

predictors to estimate performance without relying on those period grades.

5. Model Development

Two regression algorithms were trained and compared: DecisionTreeRegressor and RandomForestRegressor. The Random Forest configuration used 200 trees (n_estimators=200) and random_state=42. A separate Random Forest pipeline was trained using the early feature set, excluding G1 and G2. The fitted pipelines were saved as Joblib files in the models directory.

6. Evaluation Metrics

MAE (Mean Absolute Error): the average absolute difference between predicted and actual grades. Lower values indicate smaller average errors.
RMSE (Root Mean Squared Error): the square root of the average squared prediction error; it penalizes larger errors more heavily. Lower is better.
R^2 (coefficient of determination): indicates how much of the target variance is explained by the model on the evaluated data. A value closer to 1 indicates a stronger fit; it is not classification accuracy.

7. Results

Model	MAE	RMSE	R^2
Decision Tree	1.28	2.34	0.73
Random Forest (Full)	1.18	2.00	0.80
Random Forest (Early)	2.97	3.75	0.31

The full Random Forest achieved the best recorded test-set metrics among the three models. The early model has lower performance, reflecting the reduced information available when G1 and G2 are excluded. Five-fold cross-validation on the training data for the full Random Forest produced a mean R^2 of approximately 0.902 with a standard deviation of approximately 0.027. This cross-validation result is separate from the held-out test-set R^2 of 0.80.

8. Streamlit Prediction Application

The Streamlit application loads early_prediction_model.pkl and collects 30 input features through an interactive form. It constructs a one-row Pandas DataFrame and passes it to the saved preprocessing-and-model pipeline. The predicted value is displayed on a 0–20 grade scale and grouped into indicative performance bands. The application includes a disclaimer that predictions are estimates and are not guaranteed examination results.

9. Limitations and Responsible Use

- The Mathematics dataset contains 395 student records and may not represent all schools or student populations.
- The early model's test R^2 is 0.31, so its predictions have substantial uncertainty.
- The full model uses G1 and G2, which may not be available at an early stage; its higher score should not be compared as if it used the same information as the early model.
- Feature importance and correlation describe patterns in this dataset; they do not prove that a feature causes a grade outcome.

- Predictions should not be used as the sole basis for high-stakes educational decisions. They should support, not replace, educator judgement and student support.

10. Conclusion

The project demonstrates an end-to-end supervised machine learning workflow, from data understanding and exploratory analysis to preprocessing, model training, evaluation, model persistence and an interactive prediction interface. The full Random Forest performed best on the recorded test set, while the early-prediction model provides a more practical input scenario for the Streamlit application at the cost of lower predictive performance.

11. Project Files and Reproducibility

The project includes the dataset files in dataset/, trained model pipelines in models/, Jupyter notebooks in notebooks/, evaluation results in reports/model_comparison.csv, chart outputs in visualizations/ and the Streamlit application in src/app.py. The environment dependencies are recorded in requirements.txt. To run the app from the project root after installing dependencies, execute: `streamlit run src/app.py`.

Source note: Project-specific dataset details, implementation choices and metrics in this report are based on the development information supplied during the project conversation. Confirm them against the final notebooks and source code before formal submission.